MARKET IMPACT
& ADOPTION



Open Adapter Card (OAC)

Shared Engineering Problem Turned Industry-Wide Adoption Play

Kevin Mundt – *Fellow – Advanced Micro Devices (AMD)*

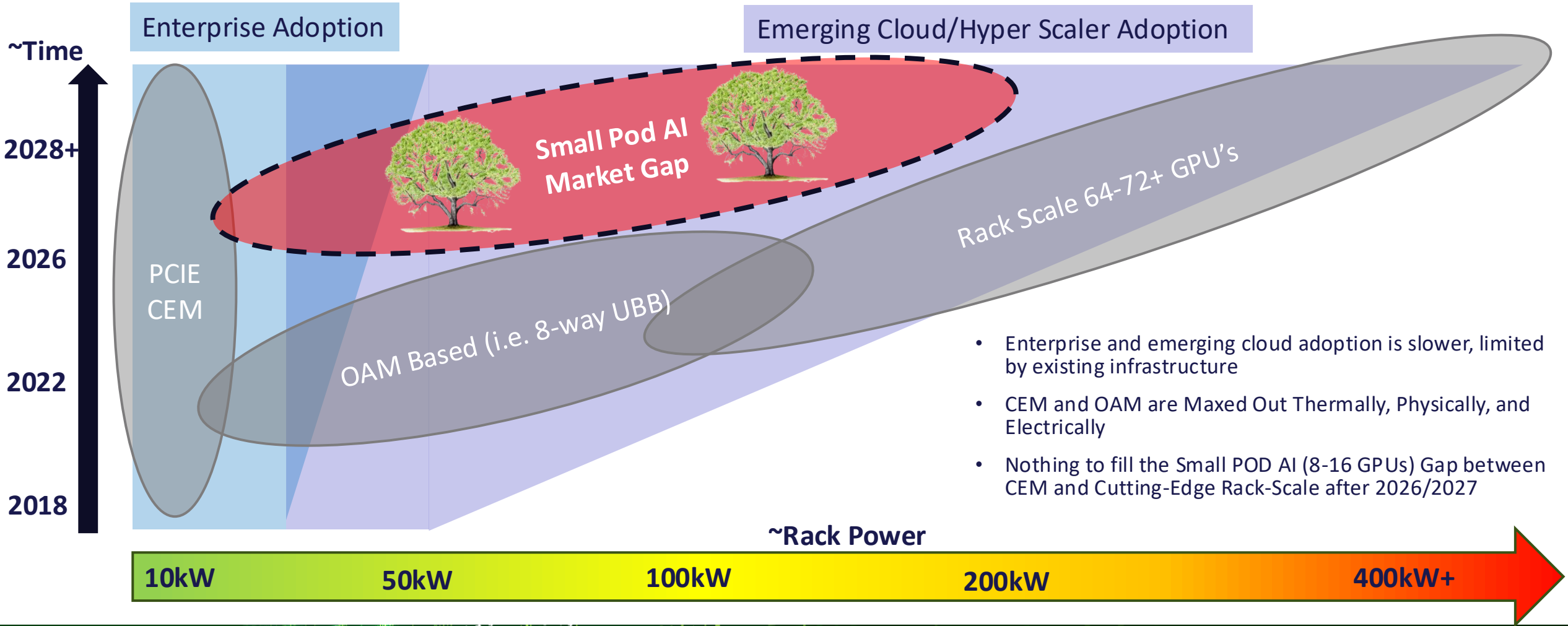
Alex Sanville – *DMTS – Dell Technologies*

OCP PLATINUM Member

*For Illustrative purposes only

The Opportunity

The Emerging and Growing Market Gap*



- Enterprise and emerging cloud adoption is slower, limited by existing infrastructure
- CEM and OAM are Maxed Out Thermally, Physically, and Electrically
- Nothing to fill the Small POD AI (8-16 GPUs) Gap between CEM and Cutting-Edge Rack-Scale after 2026/2027

Small Pod AI Market Needs

Enterprise and Emerging Cloud buyers.



Power and cooling

- Air cooling is not going away anytime soon
- Limited rack power at the site
- Fits existing enterprise facilities



Longer life cycle

- Stays in service well past a hyperscale refresh cadence
- Long-term parts and firmware support



Serviceability

- Field-replaceable parts and on-site service
- Stable hardware and firmware baselines
- Predictable uptime in production



Scalability

- Start with one pod and grow in place
- Training and inference on one footprint
- Same platform at edge and core

Enterprise Customers…

86% of racks sit at 5–30kW

77% have no plan to raise rack density

64% still run only air cooling

Why Standardize

One spec lets both sides innovate: silicon keeps its platform landing, platforms accept any silicon

THE BESPOKE TAX TODAY

Redundant Engineering	Every vendor builds the same CEM-to-rack-scale bridge
Custom Mechanicals	One-off adapters and chassis, no shared tooling
Legacy Baggage	CEM-era power and thermal limits cap every new part
Vendor Lock-in	A proprietary bridge is a single-source roadmap

WHAT ONE STANDARD UNBLOCKS

SILICON PROVIDERS
Ship new silicon knowing a platform landing exists

One standard interface

PLATFORM PROVIDERS
Ship new platforms knowing any compliant silicon fits

Decoupled roadmaps: each side innovates against a fixed interface instead of re-earning compatibility with every generation.

One Card, Unlimited Possibilities

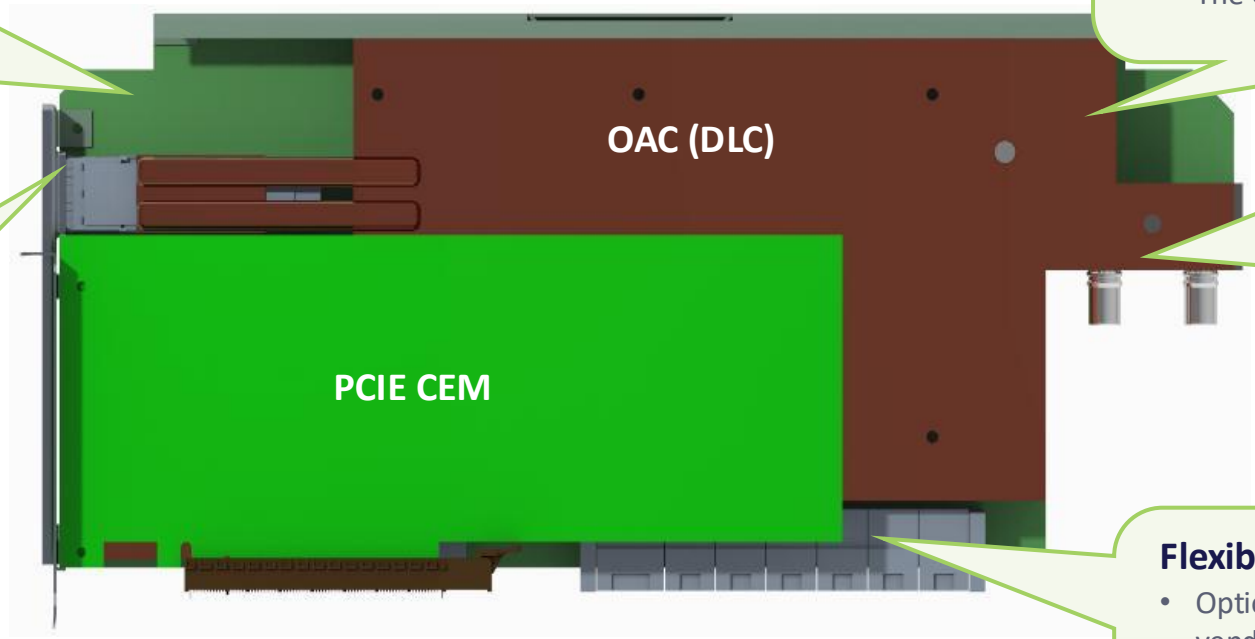
One standard, many platforms: air or liquid, field serviceable, on an enterprise life cycle

Air and DLC Variants

- Specification Defined Air and DLC Variants
- Defined with Headroom to grow

Reliability and Cost

- Vendors designing with same bag of parts increases test rigor which improves reliability
- Higher part re-use across vendors drives commodity cost down



OAC DLC Module (back) vs PCIe CEM (front): 2.2x the footprint, purpose-built headroom

Enhanced Serviceability

- No Cables needed (Power or otherwise)
- Comes with vendor defined cold plates or heatsinks
- The Card is the FRU

Standardized DLC Interoperability

- Spec defined DLC connectors ensuring one platform will work for all cards

Flexibility

- Optional Fabric connectors defined be the vendor
- Flexible connectivity and increased board size enables unlimited card and platform architectures

Proof It’s Working

This isn’t just a whitepaper. It's a spec that people are beginning to build against.

AMD intel NVIDIA Qualcomm DELL Technologies

Five companies — co-authoring OAC together

v0.9

Specification published and publicly available today

Live Design Demos

Platforms and adapters already in concept

5 Founders

Cross-industry authorship spanning silicon, systems, and OEMs

Momentum this early, from this many competitors, is the strongest signal a standard will stick.

Call To Action

- At v0.9, the spec is still being shaped — this is the moment new input actually changes the outcome
- OxMs, ODMs, and adapter vendors have real influence over the final spec.

- Specification Timeline
  - v0.9; Available Today
  - https://www.opencompute.org/w/index.php?title=Server/OpenAdapterCard

- Provide feedback
  - AMD – Greg Sellman – gsellman@amd.com
  - Qualcomm – Jared Wright – jared.wright@qualcomm.com
  - Dell – Alex Sanville – alex.sanville@dell.com
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  - Nvidia – John Norton – johnn@nvidia.com

Thank You!

